WiFi Mangoapple

Cloning WiFi Pineapples

June 22th, 2026

It ain't what you don't know that gets you into trouble.
It's what you know for sure that just ain't so.
— Mark Twain

Contents

Attempting to follow the recommendations or implement the ideas in this paper may not fit your context, may not solve your problem, and may in certain circumstances plunge your program and career into abject chaos.[1]

I. Introduction

The Hak5 Pineapple is a WiFi penetration-testing platform created by Hak5 that has become an industry standard. Its main draw is its PineAP, a rogue access point suite that enables a multitude of attacks and reconnaissance.[1][2]

While the platform and hardware are great in all, it is, however proprietary and locked to its specific hardware (or is it?[2]); that is without mentioning of course the price ($250 US as of May 2026).

This is where a Mangoapple, or pineapple cloning, comes in. Using the WiFi-pineapple-cloner project, we can 'clone' the capabilities onto cheaper hardware.[3]

II. Hardware

There are a multitude of supported routers; however, we will use the GL-MT300N-V2.

To pair with this hardware, you will also need a "Generic RT5370 WIFI adapter or MT7612U WIFI adapter".[3]

III. Build

The building process can be broken into two steps, 1) flashing OpenWrt,[4] and then 2) flashing our firmware.

A. OpenWrt

We will need to download OpenWrt, specifically *19.07.7*. You can find it at the following links, just look for `gl-mt300n-v2` under the target `ramips/mt76x8`.[5][7]

https://downloads.openwrt.org/releases/19.07.7/targets
(archive) https://archive.openwrt.org/releases/19.07.7/targets

Proceed to the web manager (default: `http://192.168.8.1`, and you may have to set up first). From there you can navigate to **Maintenance -> Update / Firmware Update**.[6] Next Upload and then flash, while ensuring that you **do *not* keep settings**.[3], [9]

B. Custom Build

Download the precompiled firmware from the project page. The build repository link can be found in the installation instructions near the bottom. Ensure you get the download for the *GL-MT300N V2* version **19.07.7**. Proceed to installing it by logging into the web interface (default: `192.168.8.1`) and proceed to **System -> Backup / Flash Firmware -> Flash new firmware image**. Next upload and flash, while ensuring that you **do *not* keep settings**.[3]

You can now navigate to `http://172.16.42.1:1471` where the control panel lives (as long as you are connected to it).

IV. Appendix

A. Debugging

- Builds General Debug
- GL.iNet Router Docs[9]
- OpenWrt: GL.iNet GL-MT300N V2[6]

a) *Stuck Booting:* If the final flash messed up (for example including keep settings), then you may be stuck with the **waiting to boot** message when you navigate to `http://172.16.42.1:1471`. To fix this you may need to re-flash your device with OpenWrt and start over, which can be done using the recovery mode.[Section IV.C]

b) *Unable to Activate PineAP:* When you are unable to activate PineAP, with the error similar to "`Monitor interface won't start!`" then you may have problems with your interface being used by another program. You can check with the command it provides (killing these may not free it up). This may also be due to the wrong interface being defined in the *PineAP* settings. To check and/or change them navigate to the *PineAP* tab and change the

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[1] It can also perform war-driving, Man in the Middle (MitM), Fake HTTPs, Evil Portal Attacks, and more.

[2] The project, mainly used to create our *Mangoapple*, actually *extracts* the original WiFi Pineapple Tetra firmware (in `build.md`).[3] Others have also taken that firmware and ported them, for example:[4].

[3] The *GL-MT300-V2* only works with the 2.4GHz spectrum with its inbuilt antennas. So specifically, we need this for both another antenna and operation on the 5GHz spectrum. [5], [6]

[4] You could get a *OpenWrt* preinstalled (for example: PrivateRouter and Amazon), instead of flashing it yourself.

[5] You can directly download it *here*

[6] You could also, force it into recovery mode[Section IV.C], use a TFTP server, or directly navigate to `http://192.168.8.1/cgi-bin/luci/admin/system/flashops` instead. [6], [8]

Monitor Interface and *Source Interface* to the one in use. You can view that information in the networking tab.

B. Internet Access

If your host device has only one network interface, plug the *Mango's* WAN port into your home router and connect to the Mango either wirelessly (may require configuration) or through the ethernet LAN port.

It is also possible to have the Mango connect to a wireless network to provide internet without ethernet.[7] This can be done by navigating to the Network settings in the *WiFi Client Mode* window. From there you can select a interface to use, scan, select, and authenticate to a wireless network as if the router was a client.

C. Recovery Mode

Recovery mode enables a web interface which allows you to manually update the firmware. This requires a direct ethernet connection, and your computers connected ethernet port IPv4 address set to `192.168.1.2/24`. You can use recovery mode instead of going through the web manager to flash the firmware.[6]

First, make sure the router is unplugged. Press and hold the reset button. Plug the router back in. Continue to hold the reset button for three to six seconds[8] and then release.[6]

Navigate to `http://192.168.1.1` which will be the web portal we will use to update the firmware. Upload the firmware, then wait for the lights to stop blinking.[6]

D. LED Indicators

The LED's on top of the router are a helpful indicator of what is happening on the router.

LED	Indicator	Description
LED 1 (Power)	Power	Power is applied
LED 2 (Customized)	Customized	This is customized LED.
LED 3 (WiFi connection)	Wireless	Indicates WiFi signal (solid) and data transfer (flashing)

Table I: Button Definitions[5]

E. Visual Resources

For those who are visually inclined:

- (youtube) Build a $23 Wi-Fi Pineapple
- (youtube) I Turned My Old Rounter Into an Ethical Hacking Tool

F. Router Outline

```
 ,-----------: Power Port
 |      ,-----: LAN Port
 |      |  ,--: WAN Port
 v      v  v
+-==----==-==-+
|             |
] <-: USB     |
[ <-: Reset   |
{ <-: Switch  |
|             |
| (.) (.) (.) |
+-------------+
   ^   ^   ^
   |   |   `--: LED 3
   |   `------: LED 2
   `----------: LED 1
```

[7]This feature is advertised for the stock firmware, but this is talking about in use with the *Mangoapple*.

[8]I had better success with six seconds. Looking at the indicator lights may also help, by waiting for them to go solid.

Bibliography

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